

Factory Production & Logistics Simulation System

Objective

Design and implement a concurrent system that simulates a factory pipeline where items are produced, processed, stored, and transported into final stock.

System Overview

You must implement a multi-stage processing pipeline consisting of the following components:

1. Machines (Producers)

- Generate items continuously or on demand
- Each item must contain:
 - Unique ID
 - Type/category (used for later grouping)
- Machines push items into the Order Line

2. Order Line (Input Buffer)

- Acts as a bounded queue between production and processing
- Has a fixed capacity
- If full, system must handle overflow (design choice: block, reject, or retry)

3. Quality Checker (Processor)

- Continuously consumes items from the Order Line
- Processes each item with a random delay (simulating real-world variability)
- Each item is randomly classified as:
 - **Passed** → sent to Storage
 - **Failed** → handled according to your design (drop, retry, or log)

4. Storage (Intermediate Warehouse)

- Stores only successfully processed items
- Items must be grouped by their type/category
- Acts as a temporary holding area before transport

5. Transport System (Ordering Car / Loader)

- Periodically arrives to collect items from Storage
- Selects items grouped by type/category

- Transfers them into final Stock
- Has constraints such as capacity and scheduling interval

6. Stock (Final Destination)

- Stores finalized items permanently
 - Organized by item type/category
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System Flow

Machines → Order Line → Quality Checker → Storage → Transport System → Stock

Functional Requirements

- Ensure correct movement of items through all stages
 - Enforce order line capacity limits
 - Simulate asynchronous processing with random delays
 - Ensure safe concurrent access across components
 - Maintain correct grouping of items in Storage
 - Allow Transport System to selectively pick items by type
 - System must run continuously without deadlocks or crashes
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Design Requirements

You must consider:

- Thread safety or async/await architecture
 - Queue or concurrent collection selection
 - Backpressure handling between stages
 - Separation of responsibilities per component
 - Data consistency in shared storage
 - Scheduling strategy for Transport System
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Optional Enhancements

- Multiple machines producing simultaneously
- Multiple order lines with load balancing

- Multiple quality checkers in parallel
- Retry mechanism for failed items
- Priority items (VIP fast-track flow)
- System metrics (throughput, latency, failure rate)